

Problem Solving and Calculations

Set 5: Moments and Equilibrium

Notes

1. What is the difference between m N and mN ?
2. Mario, a construction worker, is using a large wrench to tighten bolts joining two steel girders. He holds the wrench at a point that is 750 mm from the bolt and applies an average force perpendicular to the wrench of 160 N . Find the torque he applies to the bolt.
3. Louise uses a torque wrench to tighten the cylinder head bolts on her car. A torque wrench has a dial that allows you to preset the torque you want, then gives you a warning sound when you reach this set torque. Louise sets her wrench to warn her when she exceeds 88 N m . If the handle of the wrench is 400 mm long how much force does she need to apply to the handle before she hears the warning?

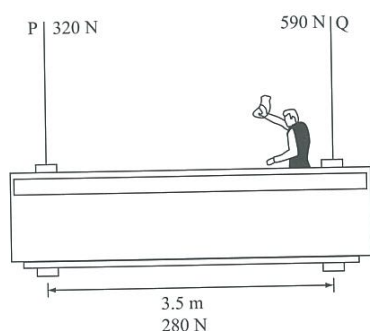


4. Explain why buses and trucks usually have much larger diameter steering wheels than normal passenger cars have, while racing cars usually have smaller diameter steering wheels than normal passenger cars have.
5. While hiking through the bush carrying a heavy backpack, Michael finds himself leaning forward as he walks. Why does he do so?
6. While playing in the park with his two children a father sits on the see-saw at a point 1.6 m from pivot point. Amanda tries to balance him by sitting at the opposite end, which is 2.5 m from the pivot.
 - a) Amanda has a mass of 24 kg and father's mass is 60 kg . Show that they cannot achieve a balance.
 - b) Ben balances the see-saw by sitting 0.50 m front of Amanda. Calculate Ben's mass.
7. When Jackie replaced the standard tyres on her sports car with a set of low profile tyres, the car seemed more powerful and was able to achieve greater acceleration. Explain how this was possible. (Low profile tyres reduce the overall diameter of a wheel.)
8. A plumber carries a long 36 kg pipe on his shoulder finds that he must pull down on the pipe with his outstretched hand 450 mm in front of his shoulder to keep the pipe balanced. He discovered later that the centre of mass of the pipe was actually 120 mm behind him. Calculate:
 - a) the downward force that his hand has to exert while he carries the pipe in this way; and
 - b) the upward force that his shoulder has to exert while he carries the pipe in this way.



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9. A uniform wooden bench type seat has a mass of 25 kg and a length of 1.9 m. The legs of the bench are 400 mm from each end. A person sits on the end of the bench, their weight acting at a point 100 mm from the end of the bench. What is the maximum mass of a person who can on the bench in this way without causing it to tip up?
10. John works on tall buildings as a window cleaner and stands on a horizontal platform suspended at its ends vertical cables P and Q. The 3.5 m long platform has mass spread uniformly along its length and weighs 280 N. On a particular occasion the tensions in the cables P and Q are 320 N and 590 N respectively.
 - a) Determine John's weight.
 - b) How far he is standing from the centre of the platform?
11. The front wheels of Janet's car support 8000 N of its weight, while the rear wheels support the remaining 7000 N. She wanted to find out how far the centre of gravity was from the front wheels, so she measured wheelbase (distance between the front and rear wheels) of her car and found it was 3.20 m. How far was her car's centre of gravity from its front wheels?
12. Ralph is building a bridge. He needs to lift a long, non-uniform tree trunk weighing 48.0 kN into a horizontal position onto two supports, using a crane with a maximum lifting capacity of 30.0 kN. If the crane operates at maximum load when it lifts the first end into position, find:
 - a) the lifting force the second operation at the other would need,
 - b) the distance to the centre of mass of the log from its heavier end.
13. Engineers design racing cars with a low centre of mass and widely spaced wheels to improve their high speed cornering ability. Explain, using moments, the physics on which they base this design.



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14. At the end of a long day of painting his two-storey house Keith finds that he can't quite reach a part of the external wall near a balcony. He decides to improvise and use a heavy plank that is about 4.0 m long and has a mass of 37.5 kg.

a) How far beyond the edge of the balcony will he be able to stand if he arranges the plank so its centre of mass is 750 mm in from the balcony edge?

Keith's mass is 73.5 kg.

b) Describe a simple way Keith could use to increase this distance.



15. A bridge has a weight of 3.15×10^5 N and two piers, 29.7 m apart, support it. The bridge's mass is uniformly distributed along its length. When a truck weighing 5.30×10^4 N is 10.7 m from the far end of the bridge and a car weighing 1.25×10^4 N follows 11.4 m behind it, find the weight supported by each pier.

16. In a science experiment several students tried to touch their toes while they stood with their backs to the wall. They found that this was not possible if their heels were touching the base of the wall and they kept their legs straight. Explain why none of the students could perform this apparently easy task.



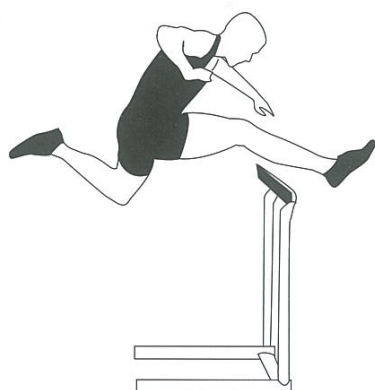
17. By inverting your body to a handstand position your centre of mass moves much closer to the ground.

A lower centre of mass should give you more stability.

Why is it therefore more difficult to remain in a handstand position than to stand upright as normal?

18. Explain why you can not avoid swaying from side to side when you try to walk along a straight line with your arms folded.

19. Good hurdlers have a distinctive action as they clear each hurdle in a race. You may have noticed that they lean their upper bodies well forward close to horizontal as they stride over each hurdle. Can you suggest why this action improves an athlete's performance in this event?



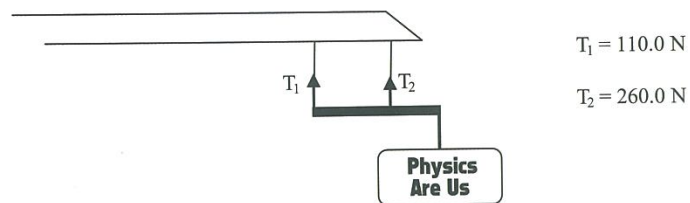
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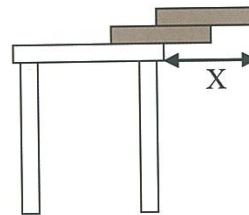
Notes

20. A uniform sign hangs from a beam outside a shop as shown in the diagram below. The beam has a mass of 30 kg, is 3.0 metres long and is suspended from the eaves of the shop by two identical wires, that are under tensions T_1 and T_2 respectively.



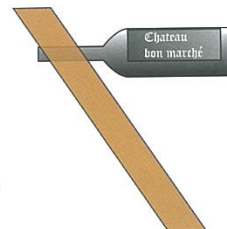
Calculate the mass of the sign and the distance along the beam where the second supporting wire is attached.

21. Two identical wooden planks, each of length 80 cm are positioned so that they hang over the end of a table as shown below.



What is the maximum possible distance, x such that the planks do not topple over?

22. Below is a diagram of a novelty one-bottle wine rack. It consists of a flat piece of wood 80 mm wide, 15 mm thick, and 200 mm long. An angled hole at one end fits the neck of a bottle. The rack, when holding a bottle, is quite stable if you stand it on a flat table as shown. Explain the critical factor in its design. Would one design suit all bottles?



23. A door, 2.40 m high and 0.90 m wide, weighs 206 N. Its centre of mass is located at its geometric centre. Hinges attached 0.30 m from its top and bottom support the door.
- Assuming that the door's weight is supported entirely by the upper hinge, find the magnitude and direction of the force that the lower hinge exerts on the door.
 - If instead each hinge supports half the weight of the door, find the magnitude and direction of the force that each hinge exerts on the door.

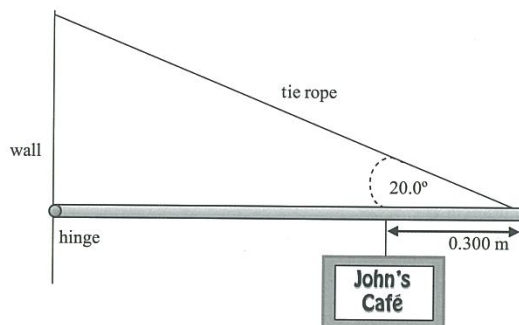
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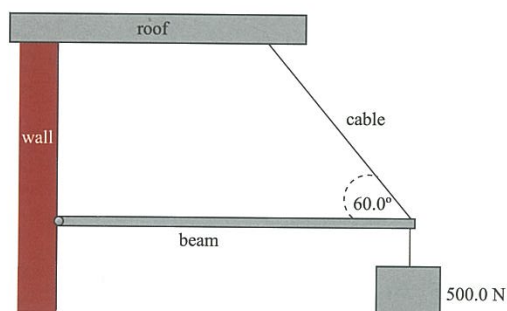
24. A shop sign has mass 45.0 kg and hangs from a uniform 12.0 kg beam as shown. The beam is 1.60 m long.



Determine

- the tension in the tie rope
- the vertical and horizontal components of the force acting on the beam at the hinge
- the magnitude and direction of the force acting on the beam at the hinge.

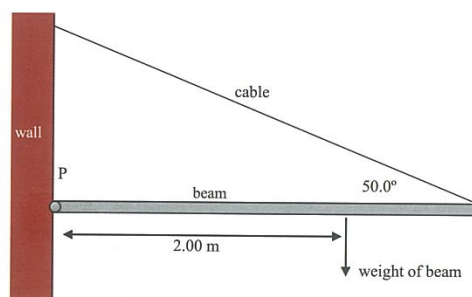
25.



The diagram shows a uniform beam, 1.40 m long and of mass 35.0 kg , that is hinged at the wall. A 500 N mass hangs from the unhinged end. This end is supported by a cable attached to the roof above.

Calculate the tension in this cable, and hence the size and direction of the force that the hinge exerts on the beam.

26. A non-uniform beam of mass 50.0 kg has its centre of mass located 2.00 metres from the pivot point P. The beam is supported by a cable connected 2.40 metres from P as in the diagram below. A man of mass 75.0 kg stands at P, then begins to walk out on the beam.



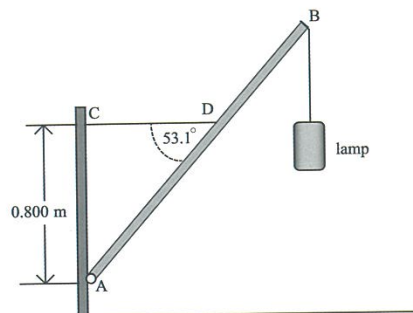
If the breaking strain of the cable is $1.36 \times 10^3\text{ N}$, how far out can he walk before the rope breaks?

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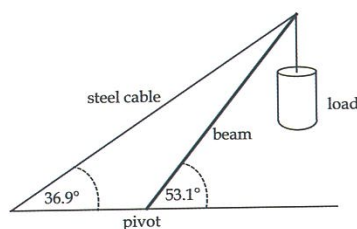
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27. A lamp hangs from the structure shown in the diagram below.



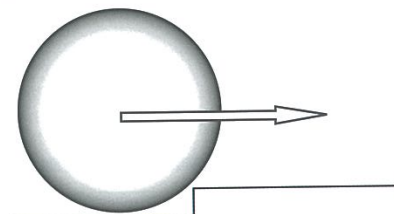
The uniform support AB is 1.70 m long and has mass 5.00 kg; the lamp's mass is 10.0 kg. Find

- the tension in the light horizontal cable CD,
 - the horizontal and vertical components of the forces that the pivot A exerts on the pole AB, and hence
 - the magnitude and direction of the force exerted by the pivot A on the pole.
28. A simple crane is made from a 2.00 tonne beam that is pivoted at the ground. Its centre of mass is one third of its length from the bottom. The beam is supported by a steel cable, and the crane has a 5.00 tonne load attached, as shown below:



Determine

- the tension in the steel cable
 - the magnitude of the reaction force exerted on the beam by the ground.
29. In order to roll a metal wheel of radius 50.0 cm up over a kerb 0.200 m high, Max has to pull horizontally at the axle with a force of 240 N.



Calculate

- the mass of the wheel
- the size and direction of the minimum force which, applied at the axle, could roll the wheel up over the same kerb.

- Set 3
- 5 0.82 m s^{-2} toward the centre of the circle
 - 6 61.9 N
 - 7 a) 1.54 m s^{-2}
 - 7 b) 17.5 N
 - 8 17.6°
 - 9 a) 691 N
 - 9 b) 3.55 s
 - 10 a) yes
 - 10 b) 3.09 kN
 - 10 c) 15.3°
 - 11 15.8 m s^{-1}
 - 12 a) 37.8 m s^{-1}
 - 12 b) $1.90 \times 10^4 \text{ m s}^{-2}$
 - 12 c) 17.7 kHz
 - 13 b) $2.59 \times 10^{-16} \text{ N}$
 - 13 d) $5.53 \times 10^8 \text{ m}$
 - 13 e) it would drop 30.6 m
 - 16 a) 88.5 m s^{-1}
 - b) 198 m s^{-1}
 - 17 87.4 m s^{-1}
 - 18 a) 180 N upward (b) 20.4 N downward
 - 19 a) 20 m
 - 20 b) 4.32 m s^{-1}
 - c) $2.00 \times 10^3 \text{ N}$
 - 21 b) 5.94 m s^{-1}
 - c) at top, 0; at bottom, 1176 N upward
 - d) at top, 441 N upward; at bottom, 735 N upward
 - 22 a) at A: 8.30 m s^{-1} ; at B: 12.1 m s^{-1}
 - b) at A: 61.6 N ; at B: 208 N
- Set 4
- 6 $6 \times 10^{24} \text{ kg}$
 - 7 $1.72 \times 10^{-6} \text{ N}$
 - 8 a) $2.64 \times 10^6 \text{ m}$
 - 8 b) 8.16 m s^{-2} toward the Earth
 - 8 c) $7.55 \times 10^3 \text{ m s}^{-1}$
 - 9 $3.80 \times 10^8 \text{ m}$
 - 11 b) $2.38 \times 10^{20} \text{ N}$ toward the Sun
 - 18 $5.74 \times 10^3 \text{ s}$ (1.59 hours)
 - 19 a) $1.37 \times 10^4 \text{ m s}^{-1}$
 - 19 b) $1.90 \times 10^{27} \text{ kg}$
 - 20 $5.97 \times 10^{24} \text{ kg}$
 - 21 $3.59 \times 10^7 \text{ m}$
 - 22 $3.05 \times (\text{radius of Moon's orbit around Earth})$
 - 23 a) $2.07 \times 10^{22} \text{ N}$; $9.20 \times 10^{21} \text{ N}$
 - 23 b) $5.37 \times 10^4 \text{ m s}^{-1}$; $4.39 \times 10^4 \text{ m s}^{-1}$
- Set 5
- 2 120 N m
 - 3 220 N
 - 6 b) 18 kg
 - 8 a) 94 N
 - 8 b) 447 N
 - 9 46 kg
 - 10 a) 630 N
 - 10 b) 0.75 m toward Q
 - 11 1.5 m from the front wheels
 - 12 a) 18 kN
 - 12 b) $0.375 \times \text{length of log from the heavier end}$

- 14 a) 383 mm from the balcony
 15 $1.86 \times 10^5 \text{ N}$; $1.95 \times 10^5 \text{ N}$
 20 7.76 kg; 2.57 m from the T_1 end.
 21 60 cm
 23 a) 137 N
 b) lower hinge 172 N toward the door at 36.9° above the horizontal
 upper hinge 172 N toward the wall at 36.9° above the horizontal
 24 a) $1.95 \times 10^3 \text{ N}$
 b) vertical = 109 N upward, horizontal = $1.83 \times 10^3 \text{ N}$ out from wall
 c) $1.84 \times 10^3 \text{ N}$ out from wall at 3.40° above the horizontal
 25 tension = $1.09 \times 10^3 \text{ N}$; force at hinge = 551 N out from the wall and 10.1° below the horizontal
 26 2.07 m from P
 27 a) 156 N
 b) vertical component = 147 N, horizontal component = 156 N
 c) 214 N to right and 43.3° below the horizontal
 28 a) $2.39 \times 10^5 \text{ N}$
 b) $2.97 \times 10^5 \text{ N}$
 29 a) 18.4 kg
 b) 144 N at 26.9° above the horizontal

Electricity and Magnetism

- Set 6 5 0.042 N m
 10 2.4 N
 12 0.075 N downward
 Set 7 1 0.57 V
 2 a) 2.3 A
 5 a) 0.21 mV
 5 b) $2.1 \times 10^{-4} \text{ Wb s}^{-1}$
 7 7.1 mV
 8 a) 39 mV
 8 d) 7.9 mA
 9 a) $8.0 \times 10^{-4} \text{ V}$
 11 225 turns
 12 a) 2.5 m s^{-1}
 12 b) $1.0 \times 10^{-4} \text{ N}$
 13 a) 83 mT
 Set 8 1 a) 60 kW
 1 b) 300 J
 1 c) 150Ω
 2 a) 14.6 m^2
 2 b) 1.25Ω
 3 a) 83 A
 3 b) 18Ω
 6 a) 0.0200 times
 6 b) 10 000 turns
 6 c) 0.0196 times
 7 a) 23Ω
 7 b) $1.3 \times 10^3 \text{ C}$
 8 a) 100 A
 10 a) 31.3 kW
 10 b) 0.31 kW
 12 a) 2.00 kW
 12 b) 0.125 m
 13 a) 6.25 kW
 15 35 km